

Grade 5 Comprehensive Mathematics Mastery & Error Remediation Exam

(Set 9)

Name: _____

Class: _____

Student ID: _____

Time: 45-50
MinsTotal: 100
Pts

Score: _____

Part 1: Conceptual Precision & Number Sense (Fill-in-the-Blanks & Comparison)

5 Questions, 20
Points

1. Quotient Invariance & Decimal Point Scaling (4 pts)

Given $48 \div 16 = 3$, write down the quotients directly using the rules of division:

(1) $4.8 \div 16 =$ _____ (2) $0.48 \div 16 =$ _____ (3) $0.048 \div 16 =$ _____

(4) $480 \div 16 =$ _____

2. Comparing Operations with Decimals Less Than 1 (4 pts)

Fill in each $\square$ with ">", "<", or "=" without detailed calculation:

(1) $6.4 \times 0.8 \square 6.4 \div 0.8$ (2) $12.5 \div 0.5 \square 12.5 \times 0.5$ (3) $0.95 \times 1.02 \square 0.95 \div 1.02$

(4) $8.4 \div 1 \square 8.4 \times 1$

3. Recurring Decimals & Remainder Cycles (4 pts)

(1) Compare: $0.714714 \square 0.\dot{7}1\dot{4}$ (Hint: expand and compare the 7th decimal place).(2) The fraction $\frac{1}{7} = 0.\dot{1}4285\dot{7}$ has a repeating period of _____ digits. The 100th digit after the decimal point is _____.

4. Generality of Algebraic Variables (4 pts)

A square has side length a cm:(1) Its perimeter is _____ cm; (2) Its area is _____ cm²; (3) When $a = 6$, its area is _____ cm².

5. Multiplicative Comparison Sentence Structures (4 pts)

(1) "How many times 15 is 45?" Equation: _____, Result: _____

(2) "A car traveled 240 km, which is 3 times the distance a bike traveled." The bike's distance is _____ km.

Part 2: Algebraic Simplification & Sign Rules (True/False & Evaluation)

3 Questions, 15 Points

6. Combining Like Terms vs. Constants (5 pts)

True or False: The algebraic expression $4x + 7 + 2x$ simplifies to $13x$. (_____)Write the correct simplification: $4x + 7 + 2x =$

Explain why:

7. Additive Inverses & Sign Properties (5 pts)

True or False: The identity $x + x - y + y = 2x - 2y$ is always correct. (_____)

Write the standard step-by-step simplification: $x + x - y + y =$

8. "Simplify First, Then Substitute" Discipline (5 pts)

Given $y = 3.5$, evaluate the algebraic expression $15y - 7y + 20$ (**Direct substitution without simplifying is strictly prohibited**).

Step 1 (Combine like terms): $15y - 7y + 20 =$

Step 2 (Substitute value): When $y = 3.5$, Value =

Part 3: Rigorous Column Arithmetic & Mixed Operations**3 Questions, 20 Points****9. Column Division with Decimal Divisors & Quotient Alignment (6 pts)**Calculate: $197.6 \div 5.2$

(1) **Pre-calculation estimate:** Round 197.6 to _____, 5.2 to _____, estimated quotient $\approx$ _____.

(2) **Column division** (clearly show shifted decimal points):

$$5.2 \overline{) 197.6}$$

Verification Check:

Is quotient an integer or decimal?

Final result =

10. Triangle Area Formula & Rigor (6 pts)A right triangle has legs of base $b = 0.36$ m and height $h = 0.25$ m.

(1) Write the triangle area formula: Area = _____

(2) Calculate the area of this triangle (show step-by-step working and units):

Working:

Answer:

11. Order of Operations with Decimals (8 pts)

(1) $0.4 \times (3.2 - 0.8) \div 1.2$

=

=

=

(2) $7.4 \times 1.3 - 4.68 \div 0.9$

=

=

=

Part 4: Word Problems & Keyword Modeling**4 Questions, 25 Points****12. Two-Step Equal Sharing & Deducting Remainder (6 pts)**

A bakery made a batch of cookies weighing 9.2 kg in total. The baker distributed them equally into 4 gift boxes, leaving 0.4 kg of cookies on the counter. How many kilograms of cookies were placed in each gift box?

Guided thinking: Total cookies actually packed into the boxes = _____ kg.

Equation & Working:

Answer:

13. Multi-Variable Matching & Unit Alignment (6 pts)

At a supermarket, apples cost \$6.8 per kg and bananas cost \$4.5 per kg. Ms. Wang bought 15 kg of apples and 24 kg of bananas for the school cafeteria. How much did she pay in total?

Equation & Working:

Answer:

14. Average Score Total Delta Model (6 pts)

The school girls' basketball team averaged 42 points in their first 6 tournament games. They then played 2 more games, raising their overall average across all 8 games to 45 points.

(1) How many total points did the team score in the last 2 games combined? (2) What was their average score in the last 2 games?

Step-by-step Working:

Answer: (1) Total points in last 2 games: _____ pts; (2) Average in last 2 games: _____ pts.

15. Sharing Pitfall & Fraction of the Whole (7 pts)

"Emma bought a large pizza for lunch and cut it into 8 equal slices. She wants to share the pizza evenly between **herself** and **three** of her best friends."

(1) How many people are sharing the pizza altogether? _____ people.

(2) What **fraction** of the whole pizza will each person receive? _____ (fraction) = _____ (decimal).

(3) How many **slices** of pizza will each person get? _____ slices.

16. Bar Model Total Length Conservation Equation (5 pts)

The bar models below show three ribbons of identical total length: Ribbon 1 is $3x$, Ribbon 2 is $x + 8$, and Ribbon 3 is $4y$.

[Tier 1] |---- x ----|---- x ----|---- x ----| (Total $3x$)
 [Tier 2] |---- x ----|----- 8 -----| (Total $x + 8$)
 [Tier 3] |-- y --|-- y --|-- y --|-- y --| (Total $4y$)

(1) Since Ribbon 1 and Ribbon 2 have equal length, set up an equation in terms of x and solve:

Equation: _____ $\Rightarrow x =$ _____

(2) Find the total length of each ribbon, set up an equation for y , and solve:

Total length = _____ dm; Equation for y : _____ $\Rightarrow y =$ _____

17. Road Paving Collaboration Problem (5 pts)

Engineering teams A and B work together to pave a highway measuring 125 km in total. Team A paves 3.5 km per day, and Team B paves 4.5 km per day. Team A worked alone for the first 6 days due to scheduling, after which Team B joined to work together. How many days will it take to finish the remaining highway after Team B joins?

Working:

Answer:

18. Traveling Towards Each Other Equation (5 pts)

City A and City B are 540 km apart. A coach and a truck depart simultaneously from City A and City B respectively, traveling towards each other. They meet after 4.5 hours. The average speed of the coach is 64 km/h. Let the truck's speed be x km/h. **Set up an algebraic equation and solve for x .**

Let the truck's average speed be x km/h.

Equation:

Answer:

19. Surplus and Shortage Algebraic Model (5 pts)

School organized Grade 5 teachers and students for a boating activity in a park. If each boat holds 4 people, 6 people are left without a boat. If each boat takes 2 more people (i.e. 6 people per boat), exactly 1 boat is completely empty. How many boats were originally rented? How many total teachers and students participated?

Let the number of rented boats be x .

Total people in Scheme 1: _____; Total people in Scheme 2: _____

Equation & Solution:

Answer:



[Answer Key & Marking Rubric] (For Teacher Reference)

Part 1 (20 pts): 1. (1) 0.3 (2) 0.03 (3) 0.003 (4) 30 (1 pt each); 2. (1) $<$ (2) $>$ (3) $<$ (4) $=$ (1 pt each); 3. (1) $<$ (2 pts) (2) 6, 8 (1 pt each); 4. (1) $4a$ (2) a^2 (3) 36 (1+1+2 pts); 5. (1) $45 \div 15$, 3 (2 pts) (2) 80 (2 pts).

Part 2 (15 pts): 6. False (2 pts); Simplification: $6x + 7$ (2 pts); Reason: Only like terms can combine coefficients, constant 7 cannot merge with x (1 pt). 7. False (2 pts); Steps: $(x + x) + (-y + y) = 2x + 0 = 2x$ (3 pts). 8. Simplify: $8y + 20$ (2 pts); Evaluate: $8 \times 3.5 + 20 = 28 + 20 = 48$ (3 pts).

Part 3 (20 pts): 9. (1) Estimate: $200 \div 5 = 40$ (2 pts); (2) Column division: quotient is integer 38 (4 pts; deduct 3 pts if misplaced decimal point). 10. (1) $S = \frac{1}{2}bh$ (2 pts); (2) $0.36 \times 0.25 \div 2 = 0.045 \text{ m}^2$ (4 pts; deduct 3 pts if forgot to divide by 2). 11. (1) $0.4 \times 2.4 \div 1.2 = 0.96 \div 1.2 = 0.8$ (4 pts); (2) $9.62 - 5.2 = 4.42$ (4 pts).

Part 4 (25 pts): 12. Guide: 8.8 kg (1 pt); $(9.2 - 0.4) \div 4 = 2.2 \text{ kg}$ (4 pts); Statement (1 pt). 13. $6.8 \times 15 + 4.5 \times 24 = 102 + 108 = \210 (6 pts). 14. (1) $45 \times 8 - 42 \times 6 = 360 - 252 = 108 \text{ pts}$ (4 pts); (2) $108 \div 2 = 54 \text{ pts}$ (2 pts). 15. (1) 4 people (2 pts); (2) $1/4$, 0.25 (2 pts); (3) 2 slices (3 pts).

Part 5 (20 pts): 16. (1) $3x = x + 8 \implies 2x = 8 \implies x = 4$ (2 pts); (2) Total length 12 dm (1 pt), $4y = 12 \implies y = 3$ (2 pts). 17. Remaining: $125 - 3.5 \times 6 = 104 \text{ km}$; $104 \div (3.5 + 4.5) = 13 \text{ days}$ (5 pts). 18. Equation: $(64 + x) \times 4.5 = 540 \implies 64 + x = 120 \implies x = 56 \text{ km/h}$ (5 pts). 19. Expressions: $4x + 6$ and $6(x - 1)$ (2 pts); Equation: $4x + 6 = 6(x - 1) \implies 2x = 12 \implies x = 6 \text{ boats}$ (2 pts); Total people: $4 \times 6 + 6 = 30 \text{ people}$ (1 pt).